Ying-Fang Jaw

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Education

University of Wisconsin-Madison Dec 2025
Master of Science in Computer Science, GPA: 3.9/4.0 *Madison, Wisconsin*

National Tsing Hua University Jan 2022
Credit Courses in Computer Science *Hsinchu, Taiwan*

- Coursework: LLMs, HPC, Artificial Intelligence, Operating Systems, Computer Networks, Data Engineering
- Teaching Assistant: Reinforcement Learning, Database Systems, Discrete Mathematics, Software Security

Experience

NeuraSignal – Software Engineer Oct 2026 – Present

Ultrasound Simulation Platform – *Python, C++, NumPy*

- Build collision and stochastic Doppler simulators in Python and C++ for a research-grade medical ultrasound simulation platform (4-engineer team), covering signal processing, numerical validation, and automated testing

Open Source – Contributor Jan 2026 – Present

KubeRay – *Go, Kubernetes, GitHub Actions*

- Traced up to **55** s of extra startup latency per Ray worker pod to a hard-coded init-container resource limit; opened a PR (with unit tests) that derives the limit from the Ray container's resources
- Fixed History Server request failures for suspended RayClusters by checking the controller's Suspended condition instead of the spec flag, eliminating a race with the log collector's final flush
- Reduced manual CI work for maintainers: merged a flaky e2e-test fix and a stale issue/PR bot, and opened a **/retest** workflow that reruns failed GitHub Actions and Buildkite jobs from a PR comment without executing untrusted PR code

Apache Airflow – *Python, TypeScript, React*

- Merged 5 PRs across Airflow's React UI and provider packages, including a UI rendering fix and a migration of connection metadata to declarative YAML
- Wrote the locale skill for Airflow's AI translation agent, defining terminology and style rules that guide LLM output

IBM – Software Engineer Intern Jul 2025 – Aug 2025

Cloud-Native Modernization – *C++, gRPC, Docker, Kubernetes, Argo CD, Linux, k6*

- Built a http bridge for a closed-source legacy CORBA/gRPC service so Kubernetes could health-check it, automating a manual validation workflow and cutting its run time by **50%**
- Cut deployment time from 30 to 5 minutes **(6x)** by replacing manual deploys with GitOps releases on Kubernetes via Argo CD, enabling blue-green deployments
- Built a k6 gRPC load-testing suite that simulates **1,000** concurrent users to pinpoint latency bottlenecks; presented the prototype and findings in a live demo

Projects

Mini LLM Inference Engine – *Python, PyTorch, CUDA, LLM Serving, Systems Performance* 2026

- Built a GPU-based LLM inference engine featuring continuous batching, paged KV cache allocation, and memory-aware request scheduling
- Designed a block-based KV cache manager with dynamic allocation, request lifecycle management, and memory-pressure handling
- Developed a reproducible benchmarking suite measuring throughput, TTFT, TPOT, and GPU memory efficiency under concurrent workloads

Skills

Languages: Go, Python, C++, Java, SQL, TypeScript/JavaScript

Infrastructure: Kubernetes, KubeRay, Ray, Docker, Argo CD (GitOps), GitHub Actions, Buildkite, Linux

Backend & APIs: gRPC, REST, FastAPI, Spring, Node.js

Data & Cloud: Apache Airflow, PostgreSQL, MySQL, AWS

Testing & AI: k6 load testing, unit/e2e testing, OpenAI API